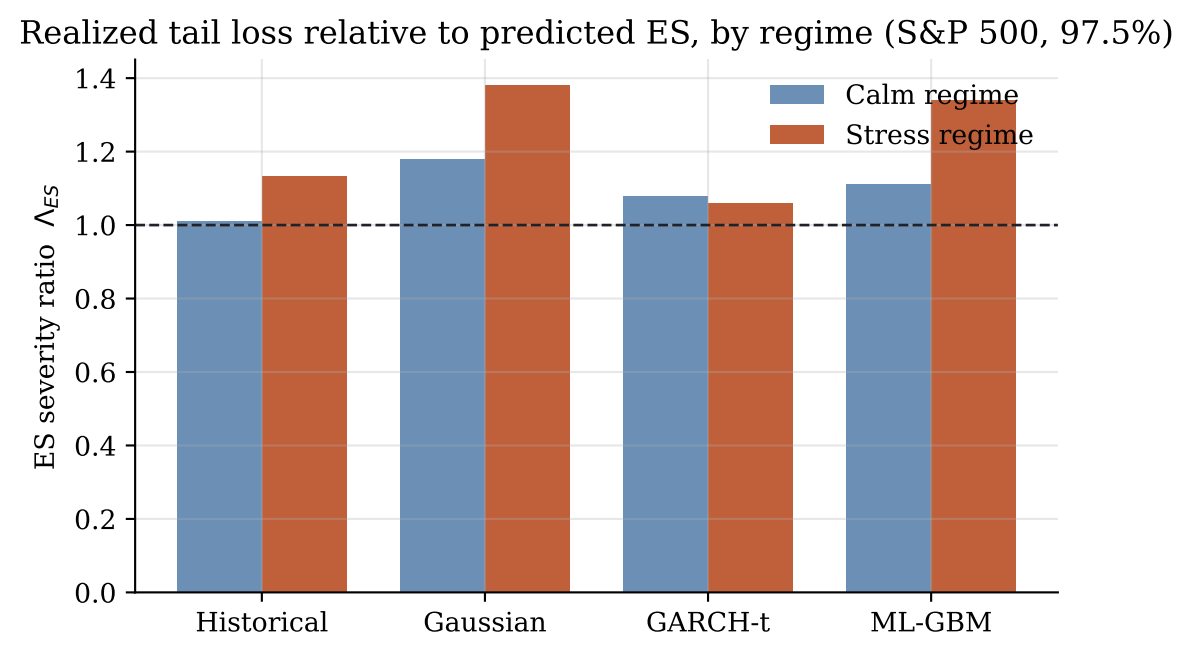
Realized tail loss relative to predicted ES, by regime (S&P 500, 97.5%)
ES severity ratio Λ_{ES}
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Calm regime
Stress regime
Historical
Gaussian
GARCH-t
ML-GBM